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Math 3108 – Linear Algebra
Exam 1
February 9, 2018

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned off or completely silenced (i.e. not on vibrate) for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 4) Failure to follow directions specific to a problem will result in the loss of points.
- 5) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 6) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 7) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 8) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 9) If you need extra room, you may use the back of the previous page. However, you must indicate you are doing so by clearly writing "BPP" on the relevant problem.
- 10) This packet has six sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 11) Once this exam begins, you will have 50 minutes to complete your solutions.

Page	1	2	3	4	5	Sum
Points Earned	6	13	16	18	12	65
Maximum Points	18	21	20	21	20	100

**DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO**

Name: _____

Multiple Choice and True/False. Circle the correct answer.

1. (8 points) Consider the matrix $A = \begin{bmatrix} 1 & 3 & 5 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$.

a) A is in _____.

(circle one)

Echelon Form

~~Reduced Echelon Form~~

Neither

b) The equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

(circle one)

True

False

c) The transformation $T(\mathbf{x}) = A\mathbf{x}$ is _____ and _____.

(circle one)

One-to-One

Not One-to-One

(circle one)

Onto

Not Onto

2. (8 points) Consider the matrix $B = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

a) B is in _____.

(circle one)

Echelon Form

Reduced Echelon Form

Neither

b) The transformation $T(\mathbf{x}) = B\mathbf{x}$ is _____ and _____.

(circle one)

One-to-One

Not One-to-One

(circle one)

Onto

Not Onto

c) The columns of B span $\mathbb{R}^4$.

(circle one)

True

False

3. (2 points) Fill in the blank. The _____ of the transformation $T(\mathbf{x}) = A\mathbf{x}$ is the set of all linear combinations of the columns of A .

a) domain

b) codomain

c) range

d) none of these

domain: $\mathbb{R}^n$
codomain: $\mathbb{R}^m$

Name: _____

Open Answer. Show ALL your work!

4. (4 points each) Let $A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 & -5 \\ 1 & -3 & 4 \end{bmatrix}$, and $\mathbf{x} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$. Compute each of the following quantities, if it exists. If it does not exist, clearly explain why.

a) AB $\begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 & -5 \\ 1 & -3 & 4 \end{bmatrix} =$

$$\begin{bmatrix} (2) - 3(1) & (0) - 3(-3) & (-5) - 3(4) \\ -2(2) + 4(1) & -2(0) + 4(-3) & -2(-5) + 4(4) \end{bmatrix} = \begin{bmatrix} -1 & 9 & -17 \\ 0 & -12 & 6 \end{bmatrix}$$

b) $A^T \mathbf{x}^T$

$$\begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} (5) - 3(3) \\ -2(5) + 4(3) \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \end{bmatrix}$$

$$5 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + 3 \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 5 - 9 \\ -10 + 12 \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \end{bmatrix}$$

c) $\mathbf{x}^T A^T$

cannot do because $\mathbb{R}^n$ of $\vec{x} \neq \mathbb{R}^m$ of A

5. (9 points) Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation. If T is the transformation that first reflects points through the horizontal x_1 axis and then reflects points through the line $x_2 = x_1$, find the standard matrix A of the transformation. (Hint: Think about what this transformation does to $\mathbf{e}_1$ and $\mathbf{e}_2$.)

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{matrix} \mathbf{e}_1 & \mathbf{e}_2 \end{matrix} \rightarrow \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Name: _____

6. (14 points) Solve the matrix equation $A\mathbf{x} = \mathbf{b}$ using augmented matrices, where

$$A = \begin{bmatrix} 1 & 3 & 1 \\ -4 & -9 & 2 \\ 0 & -3 & -6 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ -1 \\ -3 \end{bmatrix}.$$

Show all your work! State your solution in parametric vector form.

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 1 \\ -4 & -9 & 2 & -1 \\ 0 & -3 & -6 & -3 \end{array} \right] \xrightarrow{\substack{4R_1 + R_2 \rightarrow R_2 \\ -\frac{1}{3}R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 1 \\ 0 & 3 & 6 & 3 \\ 0 & 1 & 2 & 1 \end{array} \right] \xrightarrow{\frac{1}{3}R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 \end{array} \right]$$

$$\xrightarrow{-R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{-3R_2 + R_1 \rightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & -5 & -2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5x_3 \\ -2x_3 \\ x_3 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

$$x_1 = -2 + 5x_3$$

$$x_2 = 1 - 2x_3$$

x_3 is free

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

- b) (4 points) Without performing any additional row operations, state the solution set for the homogeneous equation $A\mathbf{x} = \mathbf{0}$ where A is the matrix given in part (a) above.

$$\left[\begin{array}{ccc|c} 1 & 0 & -5 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\begin{aligned} x_1 &= 5x_3 \\ x_2 &= -2x_3 \\ x_3 &\text{ is free} \end{aligned}$$

$$\mathbf{x} = x_3 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$$

7. (2 points) Fill in the blank. If the columns of an $m \times n$ matrix A span $\mathbb{R}^m$, then A has a pivot position in every column.

row

4

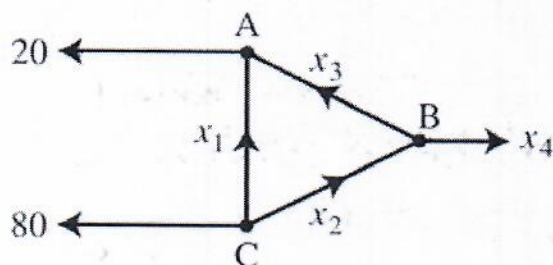
Name: _____

8. (6 points) The following *augmented* matrix in echelon form represents a system of linear equations. If the system is consistent, find the general solution in parametric vector form. Otherwise, state that the system is inconsistent and indicate why you came to that conclusion.

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 19 \\ 0 & 1 & 4 & 4 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

it is inconsistent
because this is not zero

9. (9 points) Set up an augmented matrix which could be used to find the general flow pattern of the network shown in the figure. Do not find the general flow pattern.



$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & -20 \\ 0 & 1 & -1 & -1 & 0 \\ -1 & -1 & 0 & 0 & -80 \end{array} \right]$$

flow in = flow out

A: $x_1 + x_3 = 20$

B: $x_2 = x_3 + x_4$

C: $0 = 80 + x_1 + x_2$

10. (6 points) For what values of h , if any, do the columns of $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & h \\ 3 & -4 & 5 & -7 \end{bmatrix}$ form a linearly independent set? Clearly explain your answer.

$$\xrightarrow{\frac{1}{2}R_2 \rightarrow R_2} \left[\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & \frac{1}{2}h \\ 3 & -4 & 5 & -7 \end{array} \right] \xrightarrow{R_1 + R_2 \rightarrow R_2} \left[\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & \frac{1}{2}h - 4 \\ 3 & -4 & 5 & -7 \end{array} \right] \xrightarrow{\text{swap } R_2 + R_3}$$

$$\left[\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 3 & -4 & 5 & -7 \\ 0 & 0 & 0 & \frac{1}{2}h - 4 \end{array} \right]$$

when $\frac{1}{2}h - 4 \neq 0$
 $h \neq 8$

or $h = 8$
shooodang

Name: _____

11. Consider the vectors $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix}$.

a) (2 points) Is the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ linearly independent? Yes No (circle one)

b) (4 points) List two nonzero vectors in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ other than $\mathbf{v}_1$ and $\mathbf{v}_2$, and clearly indicate how you obtained them.

$$\begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix} \times 2 = \begin{bmatrix} -8 \\ 0 \\ 2 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \times 2 = \begin{bmatrix} 6 \\ 2 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -4 & 0 \\ 1 & 0 & 0 \\ 2 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

12. Let $A = \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 4 \\ 5 & 5 \end{bmatrix}$, and $C = \begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix}$.

a) (4 points) Compute the product AB .

$$\begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix} \begin{bmatrix} 8 & 4 \\ 5 & 5 \end{bmatrix} = \begin{bmatrix} 2(8) - 3(5) & 2(4) - 3(5) \\ -4(8) + 6(5) & -4(4) + 6(5) \end{bmatrix} = \begin{bmatrix} -1 & -5 \\ 2 & 10 \end{bmatrix}$$

b) (4 points) Compute the product AC .

$$\begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 2(5) - 3(3) & 2(-2) - 3(1) \\ -4(5) + 6(3) & -4(-2) + 6(1) \end{bmatrix} = \begin{bmatrix} -1 & -7 \\ 2 & 10 \end{bmatrix}$$

13. (6 points) Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation. If $T(\mathbf{u}) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $T(\mathbf{v}) = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$, find $T(3\mathbf{u} + 2\mathbf{v})$.

$$T(3\mathbf{u} + 2\mathbf{v}) = 3 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix} + \begin{bmatrix} -2 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$$